

HOI Query

Person: holding

Object: bottle

HOI Query

Person: riding

Object: horse

HOI Query

... × 600

LLM User Message

You are an expert VDMS Optimization Engine for HOI Retrieval.

Iteration: 4 / 50 | Phase: EXPLORATION | Best Score so far: 136.1

OBJECTIVE: Score = QPS if mAP ≥ τ ($\tau = 0.15$) else 0.0 → maximize QPS subject to mAP ≥ 0.15

Parameter Physics (190-line system context — key constraints summarised):

k_neighbors ↓ → QPS ↑↑ (primary lever: k=50→295, k=100→145, k=200→58 QPS)

efSearch: QPS-neutral in VDMS batch mode · M ↑ → mAP ceiling ↑ · α : discover empirically

IVFFlat + constraint_strategy ≠ 'none' → 0 candidates returned (always infeasible Δ)

Diagnostic $g(\theta_3)$: Class = low_mAP

[mAP(θ_3) = 0.000 < 0.20 and Score(θ_3) = 0.0 < 0.80 × Score*₃ = 108.9]

⇒ IVFFlat returned mAP=0.000 (infeasible). Return to FaissHNSWFlat. Increase k or M to recover mAP ≥ τ .

History $\mathcal{H}_3$ (complete ordered trajectory — all iterations, no truncation):

iter01: HNSW M=64 efS=64 k=100 α =0.65 cs=object refs=10+centroid

→ Score=136.1 mAP=0.209 QPS=136.1 ✓

iter02: IVFFlat nlist=128 nprobe=8 k=100 α =0.65 cs=none refs=10+centroid

→ Score= 0.0 mAP=0.000 QPS=2267.6 × (mAP < τ — infeasible)

iter03: IVFFlat nlist=512 nprobe=64 k=100 α =0.30 cs=none refs=5+first

→ Score= 0.0 mAP=0.000 QPS=2281.6 × (mAP < τ — infeasible)

Untried-value hints U_t :

Active during Exploitation & Fine-Tuning only (suppressed in Exploration phase).

Each θ_{n+1} checked against $\mathcal{H}_t$ before evaluation; duplicates trigger resubmission or random fallback.

MiniMax-M2.1 via OpenRouter

Structured JSON Response — Iteration 4:

```
{ "engine": "FaissHNSWFlat",
  "params": { "M": 64, "efConstruction": 200, "efSearch": 64 },
  "k_neighbors": 50, "alpha": 0.70, "n_refs": 10, "ref_strategy": "centroid",
  "constraint_strategy": "object", ← PMGD graph pre-filter (new field)
  "reasoning": "k=50→QPS=295; M=64+cs=object keeps mAP≥τ=0.15." }
```